

Tide retrospective:
generic- (k_1, k_2) 2D unnormalised numerator

Daniel Murfet (with Claude Opus 4.7)

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Abstract

Companion to the just-landed kthkth-moment-asymptotic tide; closes the second of the two strict-improvement follow-ups named in the kthkth-partition-asymptotic retrospective. Lifts the V1 (2D unnormalised numerator) machinery from $(k_1, k_2) = (2, 3)$ to parametric (k_1, k_2) with $k_1, k_2 \geq 1$. Build clean first-try; ~ 135 lines. Mirror-shape exemption applied (no GPT consultation).

1 Setting

This tide is the natural pair to the kthkth-moment-asymptotic tide. The kthkth-partition retrospective (this morning) named two follow-ups: the moment analogue (Z2 generic) and the numerator analogue (Z1 generic). The moment tide landed earlier in this session; the numerator tide closes the pair.

2 The thing formalised

For $k_1, k_2 \geq 1$, $j_1, j_2 : \mathbb{N}$, and $t > 0$:

$$\iint x^{2j_1} y^{2j_2} e^{-tL(x,y)} dx dy = C'(k_1, k_2, j_1, j_2) t^{-(2j_1+1)/(2k_1)-(2j_2+1)/(2k_2)}$$

on the 2D separable monomial potential, where C' is the natural generalisation of the quartic-sextic numerator prefactor.

Plus the rescaled-Tendsto and Asymptotics.IsEquivalent packagings.

3 Proof strategy

Identical to the kthkth-moment tide's, with one difference: the 2D integral here is the unnormalised numerator, so `integral_separable_addSeparable` supplies the factorisation directly (no Gibbs normalisation). The 1D building block is then `kth_moment_even` (not the Gibbs-normalised `gibbsExpectation_kthPotential_even`), and the rest is `Real.rpow` algebraic rearrangement.

4 Roadblocks and lessons

No roadblocks. Build clean first-try.

Lesson: the moment / numerator pair of mirror-shape lifts land symmetrically when both have a corresponding 1D base case in the seabed.¹ Both lift cleanly to 2D via the separable factorisation machinery from `TwoD/AddSeparable.lean`.

5 Follow-ups

- **Cross-repo promotion candidate.** The trio of (partition, numerator, moment) lifts on separable monomial potentials is now a closed pattern, ready for promotion to `lean-common` once a second-repo consumer materialises.
- **The `kthkth` arc is now complete** (modulo the odd-moment analogue, which trivially vanishes by parity in either coordinate).

¹The moment tide uses `gibbsExpectation_kthPotential_even`; this tide uses the 1D unnormalised numerator `kth_moment_even`.